

Geog4/6300: Lab 0–Tornadoes and social vulnerability

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This “lab” assignment provides an opportunity to learn the basic mechanics of Github classroom, loading data, and doing some basic data manipulation using the tools in the tidyverse. Through this lab, you will identify counties with the highest levels of social vulnerability that also had high numbers of severe tornadoes (rated F3 and higher) during this study period.

This lab assesses the following learning standards:

1. Classify variables by measurement type (nominal, ordinal, interval, ratio) and justify those classifications with reference to the characteristics of the data. (Task 2)
2. Load tabular and spatial data into a code-based environment from multiple sources, including local files, remote APIs, and spatial file formats. (Task 1)
3. Filter, aggregate, and transform datasets using grouping and summary operations to answer specific analytical questions. (Task 3, 4, and 5)
4. Join multiple datasets using appropriate join strategies and explain how different join types affect the resulting output. (Task 6 and 7)
5. Use Git and Github to create and share project materials in a repository format. (Lab submission)

Loading the data

We’ll be combining two datasets for this assignment: NOAA’s database of tornadoes from 1950-2026 (so far) and the CDC’s Social Vulnerability Index (SVI):

<https://www.atsdr.cdc.gov/place-health/php/svi/index.html>

You can load the tornado data (`noaa_stormevent_tornado_2026_08.csv` in the folder `data/stormevents/`) using the `read_csv` function. The SVI data is located at `data/SVI_2022_US_county.csv` . Load the tornado dataset into an object called `tornadoes` and the SVI dataset into an object called `svi` .

(Side note: you can see how we downloaded this data directly from NWS in the “stormdata_download” script in the data folder.)

Task 1: Load the tornado data and the SVI data using `read_csv`.

```
tornadoes <- read_csv("data/stormevents/noaa_stormevent_tornado_2026_08.csv")
```

Warning: One or more parsing issues, call ``problems()`` on your data frame for details
e.g.:

```
dat <- vroom(...)  
problems(dat)
```

Rows: 80996 Columns: 52

— Column specification —

Delimiter: ","

chr (25): STATE, MONTH_NAME, EVENT_TYPE, CZ_TYPE, CZ_NAME, cty_fips, WFO, BE...

dbl (24): BEGIN_YEARMONTH, BEGIN_DAY, BEGIN_TIME, END_YEARMONTH, END_DAY, EN...

lgl (3): MAGNITUDE_TYPE, FLOOD_CAUSE, CATEGORY

❗ Use ``spec()`` to retrieve the full column specification for this data.

❗ Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
svi <- read_csv("data/SVI_2022_US_county.csv")
```

Rows: 3144 Columns: 158

— Column specification —

Delimiter: ","

chr (7): ST, STATE, ST_ABBR, STCNTY, COUNTY, FIPS, LOCATION

dbl (151): AREA_SQMI, E_TOTPOP, M_TOTPOP, E_HU, M_HU, E_HH, M_HH, E_POV150, ...

❗ Use ``spec()`` to retrieve the full column specification for this data.

❗ Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

Task 2: *Pick two variables from either dataset that use different levels of measurement (nominal, ordinal, interval or ratio). Explain which level best describes each variable and why. Also identify one variable that would be more efficiently read as a factor, and explain why.*

In the tornado dataset, the "STATE" column is Nominal data because it is just the name of the state the tornado occurred in. It can't be ranked, put in order, or multiplied/divided. Also in the tornado dataset, the "DEATHS_DIRECT" dataset is ratio data, because the amount has a definite 0, and a value can be multiplies (50 deaths is double 25 deaths). The TOR_F_SCALE, which is the tornado rating, would be ordinal data since it rated 0 through 5,

but 4 isn't twice as strong as 2. Since it can be ranked though, I think that data could be better served as a factor. I think this because it has finite levels, ranging from F (or EF) 0, to F/EF-5.

Filtering and/or summarizing the data

Open up the tornado data frame so you can look at it. There's multiple variables here related to the timing of the tornado, its location, its magnitude, and the number of injuries and fatalities. You can open the "Storm-Data-Bulk-csv_Format" pdf in the data folder to learn more about them.

For this part of the lab, you want to count the number of tornadoes rated F3 or higher (the `TOR_F_SCALE` variable) within each county (the `cty_fips` and `CZ_NAME` variables). Your resulting data frame should have one row per county and state combination, with a variable that stores the total number of severe tornadoes.

To do so, you'll need to:

1. Filter the `TOR_F_SCALE` variable to only include observations rated F3, F4, or F5.
2. Use `group_by()` to group the data by the county and state fips codes and names (`STATE_FIPS`, `STATE`, `cty_fips`, and `CZ_NAME`).
3. Use `summarise()` to count the number of these severe tornadoes within these groups.

Task 3: *Filter the storm data to include tornadoes rated F3 or higher and then count the number of events by county and state.*

```
tornadoes_F3_plus<-tornadoes %>%  
  filter(TOR_F_SCALE=="F3" | TOR_F_SCALE=="F4" | TOR_F_SCALE=="F5" | TOR_F_SCALE=="EF")  
  
County_and_State<-tornadoes_F3_plus %>%  
  group_by(STATE_FIPS, STATE, cty_fips, CZ_NAME) %>%  
  summarise(count=n())
```

``summarise()`` has regrouped the output.

- Summaries were computed grouped by `STATE_FIPS`, `STATE`, `cty_fips`, and `CZ_NAME`.
- Output is grouped by `STATE_FIPS`, `STATE`, and `cty_fips`.
- Use ``summarise(.groups = "drop_last")`` to silence this message.
- Use ``summarise(.by = c(STATE_FIPS, STATE, cty_fips, CZ_NAME))`` for per-operation grouping (``?dplyr::dplyr_by``) instead.

You also want to filter the social vulnerability data so that it only includes counties with a population over 25,000 people. This allows us to only track counties with a moderate or large population. Use the SVI Documentation pdf in the data folder to determine which variable this is, recognizing that you want the **estimate** and not the margin of error (MOE).

Filter the data below.

Task 4: *Filter the SVI data to include counties with a population of 25,000 or more.*

```
Pop_Above_25k<-svi %>%  
  filter(E_TOTPOP > 25000)
```

Let's add one more filter based on social vulnerability. When you look at the documentation there are a number of "dummy" variables (0/1) that flag counties that are at or above the 90th percentile for a number of factors. For example, `F_DISABL` flags counties in the top 10% for percentage of persons with a disability. There are also cumulative flags: `F_THEME2` sums all the flagged variables within the "Household Characteristics" category.

Pick one of those flags. It could be for a specific variable or a cumulative one. Your selected flag should have a plausible connection to tornado preparedness, evacuation, sheltering, recovery, or access to warnings.

Filter the dataset you created in Task 4 so that it only includes counties you are considering vulnerable. Then list that flagged variable and explain how you chose it and (if the flag is cumulative) how you picked a numeric cutoff.

Task 5: *Filter the data from Task 4 to only vulnerable counties and explain your decision.*

```
Vulnerable_Pop_Above_25k<-Pop_Above_25k |>  
  filter(F_MOBILE==1)
```

Describe your variable, why you chose it, and (if applicable) why you chose the cutoff you did.

I chose the `F_MOBILE` variable, which is the counties with the 90th percentile of percentages of mobile homes within a county. I chose this variable because mobile homes are able to be completely destroyed with even a lower rated tornadoes, which putting them at risk, even if they take shelter in their home. Counties with a large percentage of these mobile homes are at a greater potential for higher fatalities.

Connecting the data

Next you'll need to join these tornado counts to SVI population data using the `inner_join()` function. To do so you'll need to have two fields with the same name in each dataset. The FIPS codes are in both datasets, but they have different names (`FIPS` and `cty_fips`). The following code creates a new variable called `cty_fips` in the SVI data using `rename()` .

NOTE: You may need to change the name of the population data below to match the object you used for the SVI data.

```
Vulnerable_Pop_Above_25k <- Vulnerable_Pop_Above_25k %>%  
  rename(cty_fips = FIPS)
```

Call the function above to rename your data. Now you're ready to join the data.

Task 6: Use `inner_join()` to connect the filtered tornado data to your county data from Task 5.

```
tornadoes_F3_plus<-tornadoes_F3_plus %>%  
  select(cty_fips)  
  
Tornadoes_Vulnerable_Counties<-Vulnerable_Pop_Above_25k |>  
  inner_join(tornadoes_F3_plus)
```

```
Joining with `by = join_by(cty_fips)`
```

Task 7: Find some documentation on the `inner_join()` function online or using help in R. Describe how it works, and explain how the results would have been different if you used `full_join()` instead.

```
?inner_join
```

```
starting httpd help server ... done
```

Inner joins take the data within the rows in dataset A and join them to the data in rows from dataset B. Any matching data from dataset A will be kept, and anything that does not have a matching value in dataset B will be dropped from the join and not transferred. In a full join,

everything from both dataset A and B, even if there is not matching data, is joined, with the unmatched data from dataset B still attached to the joined dataset.

Answering your research question

Task 8: *Open up your joined dataset. Which three vulnerable counties have the highest number of severe tornadoes and how many were in each? Are there cities or other notable geographic features located in these counties?*

```
Tornado_Vulnerable_Counties_Count<-Tornadoes_Vulnerable_Counties %>%  
  count(LOCATION)
```

Walker County, Alabama had the most with 11 total F/EF-3+ Tornadoes. De Soto Parish, Louisiana has the 2nd most with 9 total F/EF-3+ tornadoes. Copiah County, Mississippi had the 3rd most tornadoes with 7 F/EF-3+ rated tornadoes. Walker County, Alabama does not have any major cities, but Jasper, the largest town in the county, has over 14,500 people. Walker County is also close to Birmingham, AL. De Soto Parish, LA also does not have a major city, with Mansfield, LA being the most populated at just under 5,000 people. It is, however, close to Shreveport, LA. Lastly, Copiah County, Mississippi also does not have a major city within it, with their largest towns both under 5,000 people each. It does follow the trend of the previous 2 counties in the sense that it is located close to a major city, in this case Jackson, MS.

Challenge question

Find a peer-reviewed article published in the last ten years that uses this NWS/NOAA tornado dataset. What research question was it trying to answer? What methods were used? What are the most notable findings? Give the full citation and a summary of at least 100 words below.

Paper: Assessing United States County-Level Exposure for Research on Tropical Cyclones and Human Health

Citation: Anderson, G. B., Ferreri, J., Al-Hamdan, M., Crosson, W., Schumacher, A., Guikema, S., Quiring, S., Eddebuettel, D., Yan, M., & Peng, R. D. (2020). Assessing United States County-Level Exposure for Research on Tropical Cyclones and Human Health. *Environmental health perspectives*, 128(10), 107009. <https://doi.org/10.1289/EHP6976>

Response to challenge question: The study aimed to answer questions regarding how hazards associated with tropical cyclones, those being wind, rain, flood, and tornadoes, impacted the United States exposure at a county level. The data used included the NOAA

Storm Events Database and HURDAT for the tropical cyclone track, as well as R for the coding software. First the study defined that any tropical cyclone track within 250 km from a U.S. county would be considered, then they defined the distance away from the tropical cyclone center and timing range around the landfall for each hazard, which generally ended up being 500 km away and within 2 days of the tropical cyclone track's closest approach. Rain only varied from the timing methodology, as it was 2 days before and 1 day after the storms closest approach to properly account for rainfall. Wind also varied because HURDAT defines the area around the storm's center that received high winds of 34 knots or greater. Some notable results were that rain was the hazard that had the most exposed counties (both per year IQR and in single storm high count), while tornadoes had the least exposed counties. Another notable result is that hazards range from local issue to widespread problem. They show that tornadoes are mostly localized to a small number of counties, mostly in Florida, while rainfall impacts almost every county from Texas to Maine.

Final submission stuff

Disclosure of assistance: Besides class materials, what other sources of assistance did you use while completing this lab? These can include input from classmates, relevant material identified through web searches (e.g., Stack Overflow), or assistance from ChatGPT or other AI tools. How did these sources support your own learning in completing this lab?

I used Google and Bing to find out what errors meant, and if I was missing any key characters like pipes, which I forget frequently. These tools guided me so that I was not scrolling my code for hours trying to find an error that was minimal and just part of the learning curve of using a new software. Furthermore, error outputs can be confusing, so explanations on what they meant in simple terms helped me solve my coding errors quicker.

Lab reflection: How do you feel about the work you did on this lab? Was it easy, moderate, or hard? What are the biggest things you learned by completing it?

In general, it was pretty easy. I did forget the pipes enough to cause slowdowns but I think as I learn this software more I will get better at it. I learned about the basics of data wrangling in R, and how it can simplify efficiently handle large datasets.

That's it! When you're done with this lab, use the Render command in Quarto to create a GitHub markdown document. Then push it to GitHub using the procedure outlined in this week's videos.